Credit EDA Assignment

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Overview

- Introduction
- Data Understanding
- Data Cleaning and Manipulation
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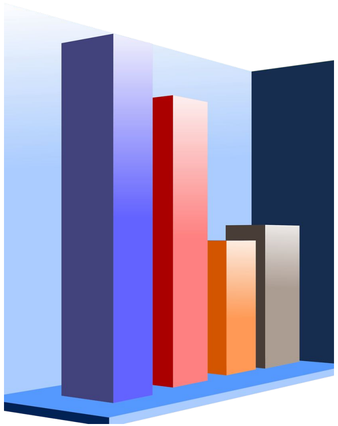
- Conclusion



Introduction

- This case study aims to identify patterns that indicate potential loan defaults, helping lenders make informed decisions.

- By using EDA, the goal is to ensure that creditworthy applicants are approved, while risky ones are identified for further action.

Data Understanding

- The application data contains 307,511 rows and 122 columns, while the previous application data has 1,670,214 rows and 37 columns.

- Both datasets have over 30% null values.

- Additionally, some "Days" columns contain negative values, and the "Amount" columns should be grouped for better clarity and analysis.

- The datasets also exhibit some data imbalance, which should be addressed.

Data Cleaning and Manipulation

- Remove Columns with High Null Values: Columns with more than 30% null values have been removed to ensure the quality of the analysis.

- Drop Non-Impactful Columns: Columns such as WEEKDAY_APPR_PROCESS_START and HOUR_APPR_PROCESS_START, which indicate when the loan process began, do not significantly impact our analysis and should be removed rather than imputed. Imputing large amounts of data could skew the results.

Data Cleaning and Manipulation

- For smaller amounts of missing data, imputation is preferred:
  - Use the mean or median for numerical columns (e.g., AMT_REQ_CREDIT_BUREAU_X).
  - Use the mode for categorical columns (e.g., NAME_TYPE_SUITE).

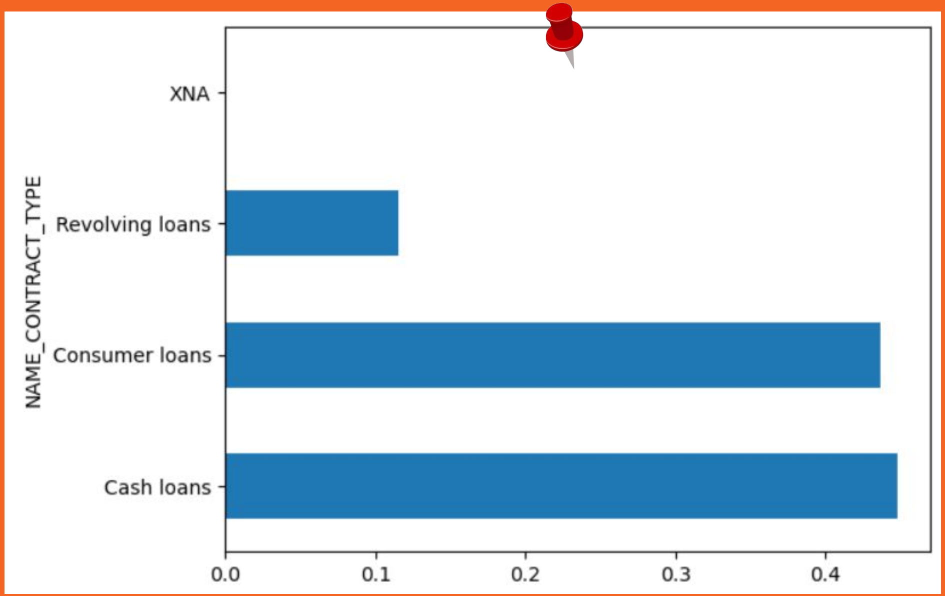
- Format DAYS_X Columns: Convert negative values in the DAYS_X columns to positive values for consistency.

- Cap Outliers in AMT Columns: Address the presence of outliers in the AMT columns by capping them for a more accurate analysis.

After understanding and cleaning the data, conducting **Data Analysis** is essential to gain insights and make informed decisions.

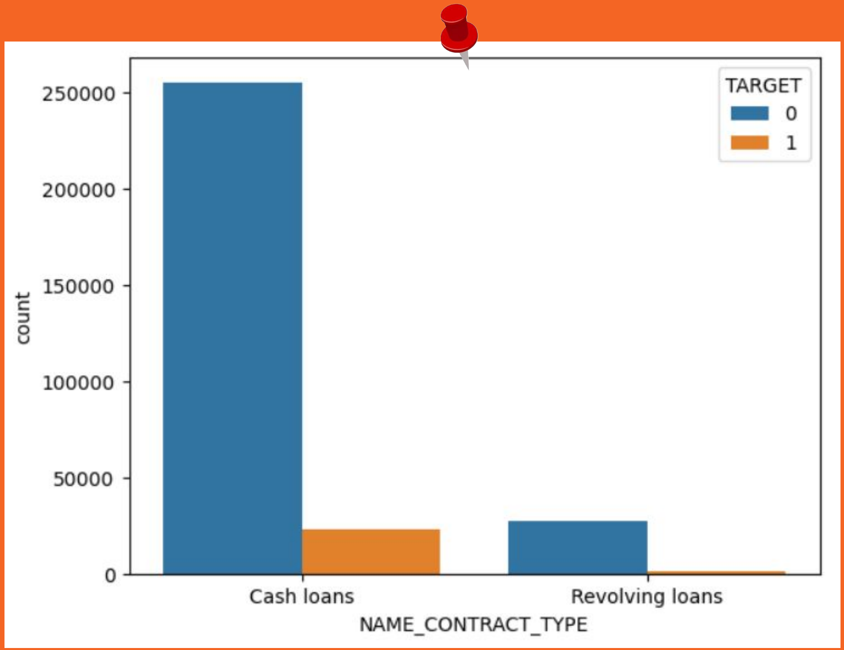
Univariate Analysis

- Cash loans and consumer loans are taken more frequently than revolving loans.



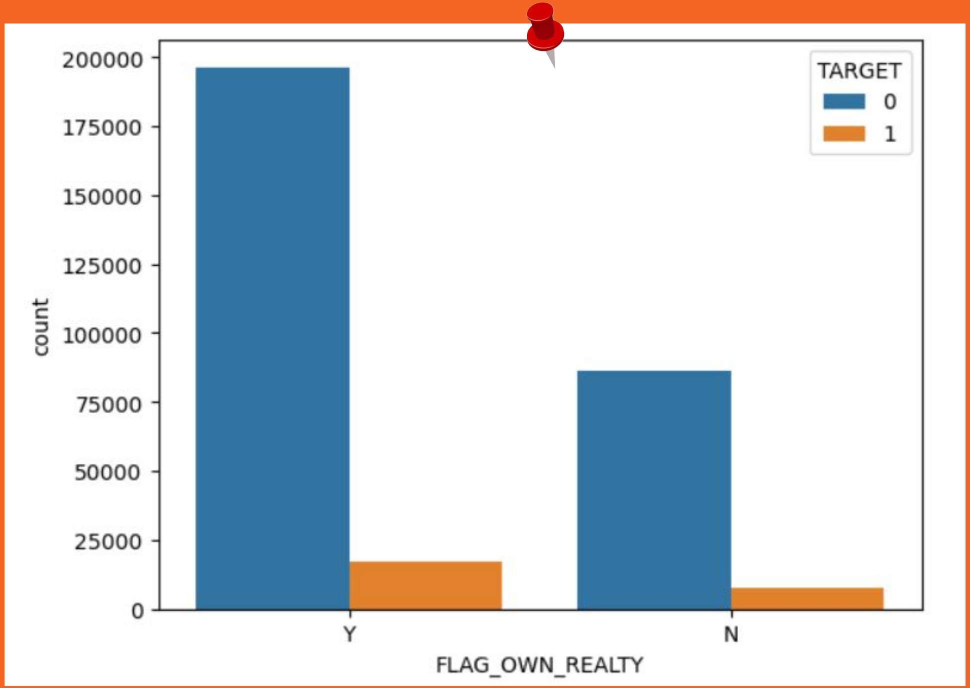
Univariate Analysis

- Cash loans have a higher default rate compared to revolving loans, yet the repayment amounts for cash loans are greater.



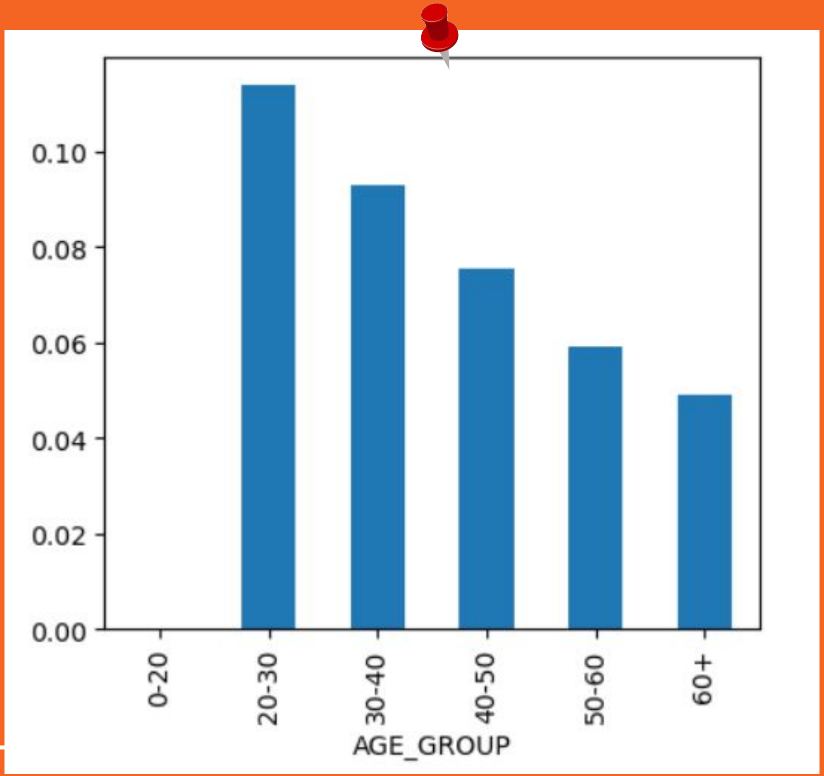
Univariate Analysis

- People who own their homes or apartments are more likely to repay their loans.



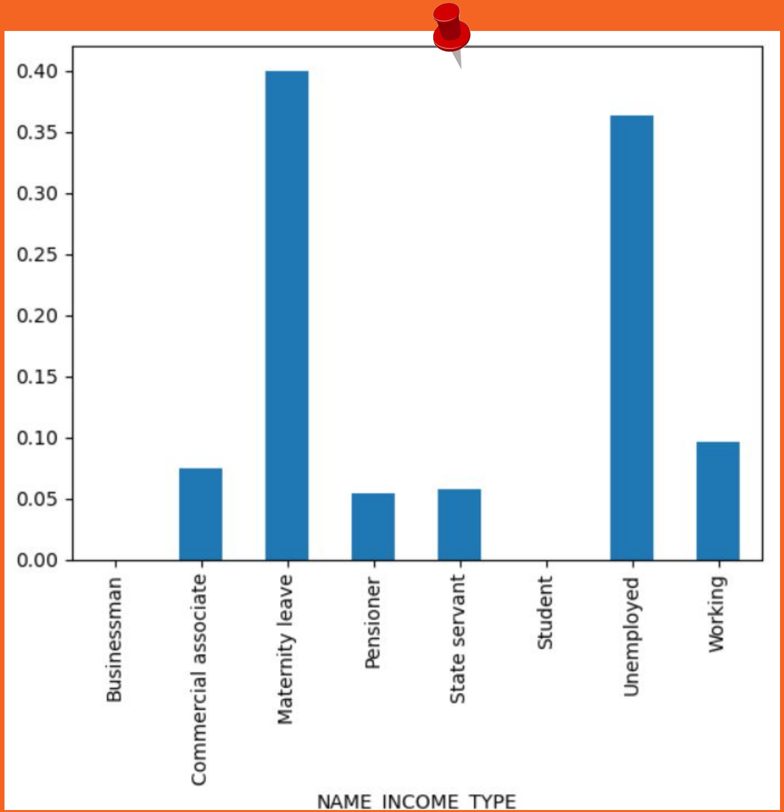
Univariate Analysis

- Individuals in the 20 to 30 age group are less likely to repay a loan, while those over the age of 50 have a higher likelihood of repayment.



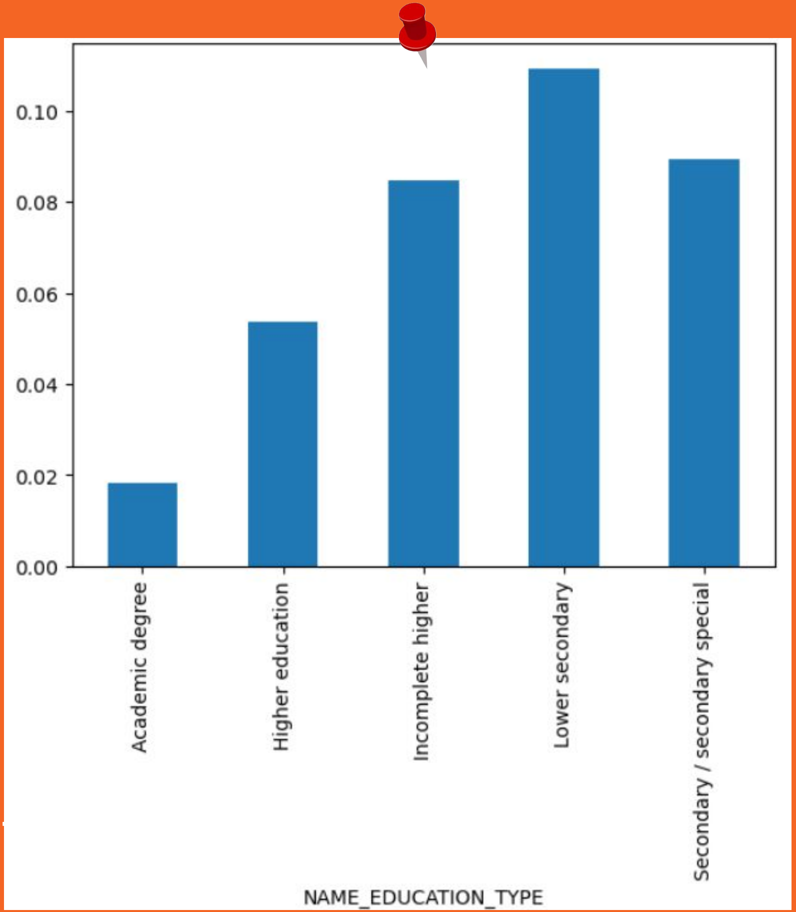
Univariate Analysis

- Unemployed individuals and those on maternity leave are more likely to default on their loans, whereas business owners and students have higher repayment rates compared to others.



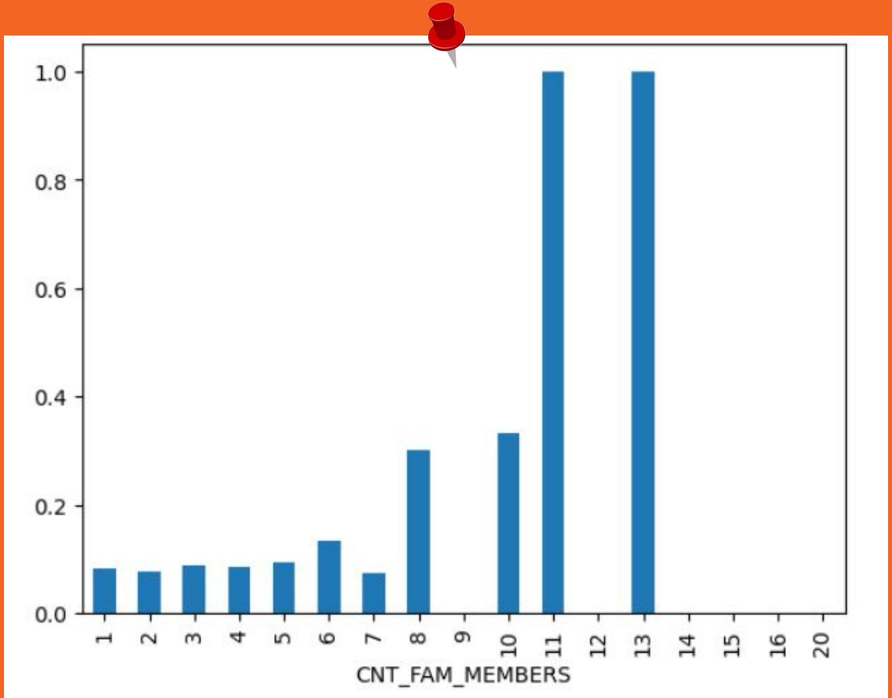
Univariate Analysis

- People with only a lower secondary education are more likely to default on their loans, while those with academic degrees have a higher likelihood of repaying their loans compared to others.



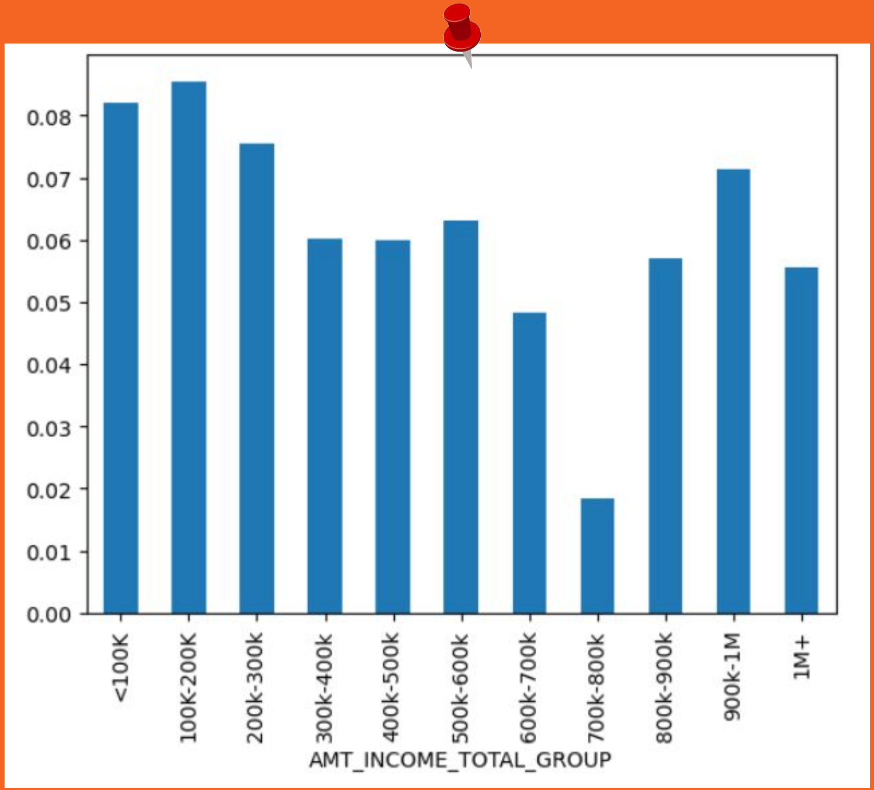
Univariate Analysis

- People with a family size of more than seven are more likely to default on their loans, while those with fewer children are more likely to repay their loans.



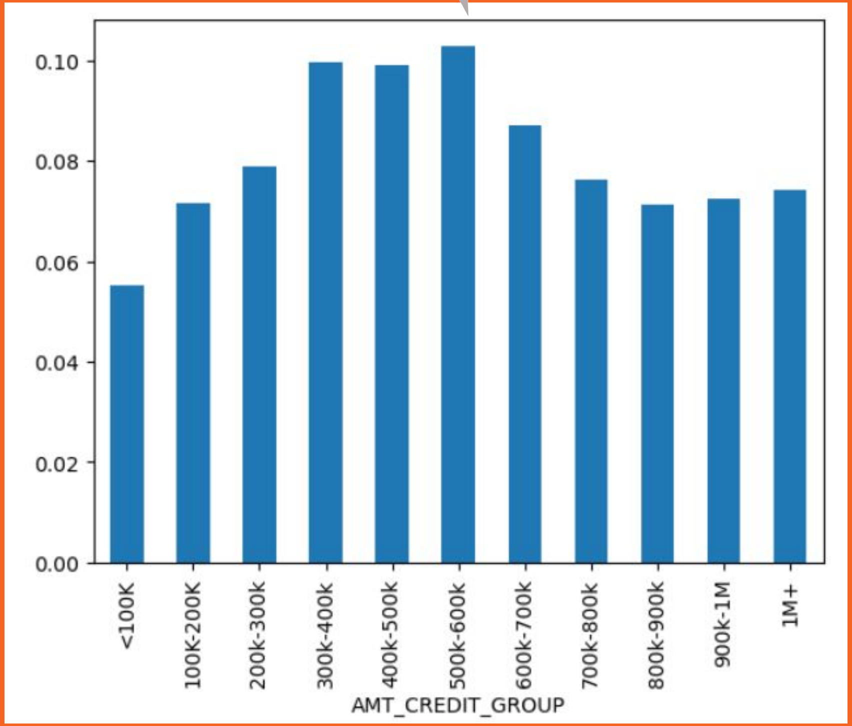
Univariate Analysis

- Individuals with an income below 200k are more likely to default on their loans, while those with an income between 700k and 800k have a higher repayment rate.



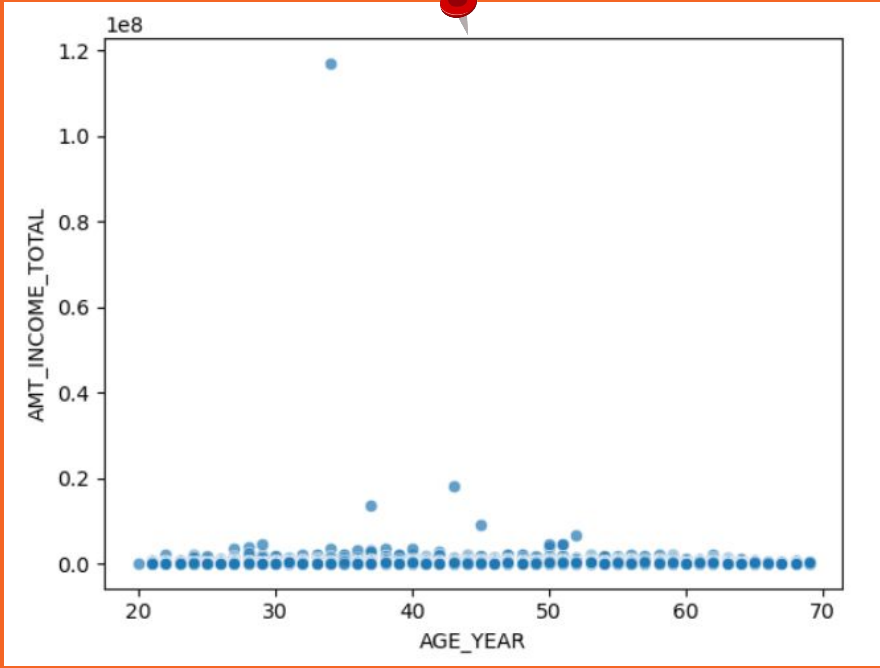
Univariate Analysis

- People who took loans between 500k and 600k are more likely to become defaulters.



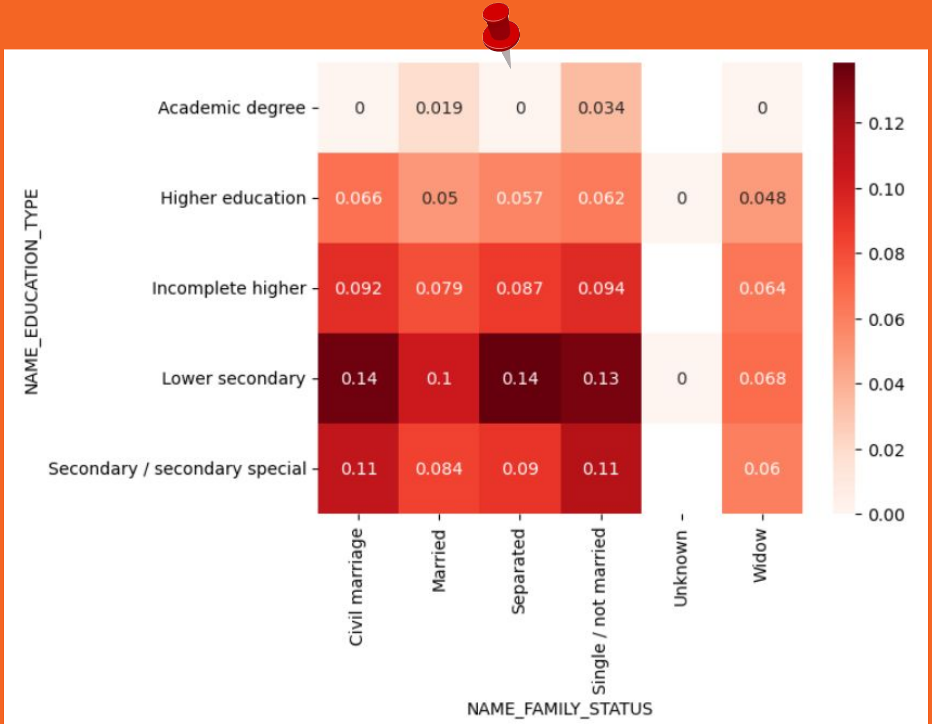
Bivariate Analysis

- There isn't a significant relationship between age and income.



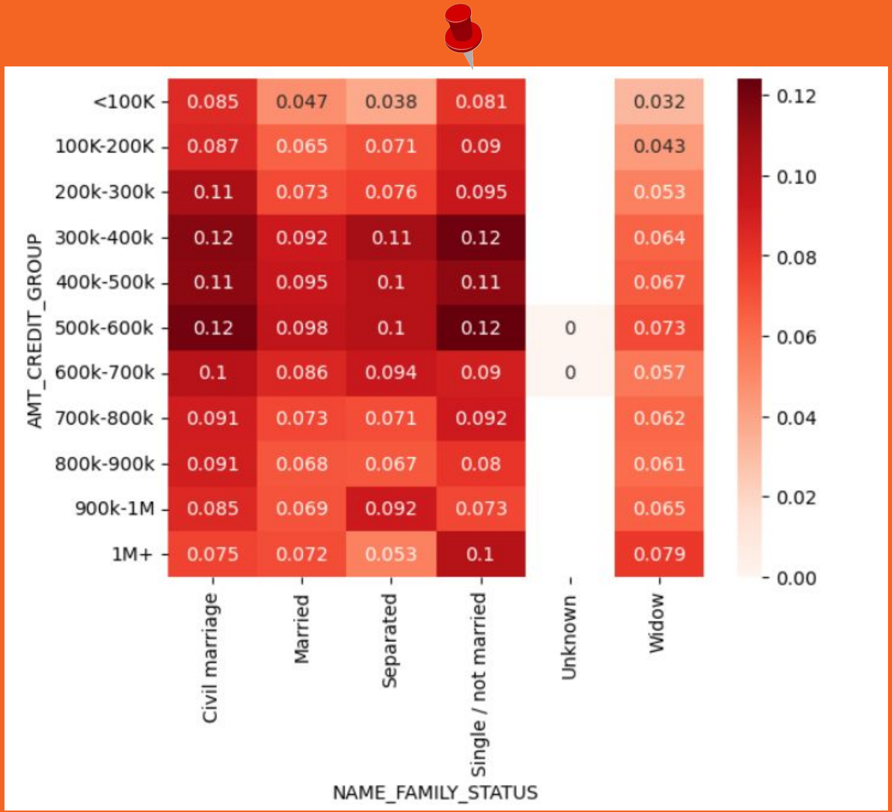
Multivariate Analysis

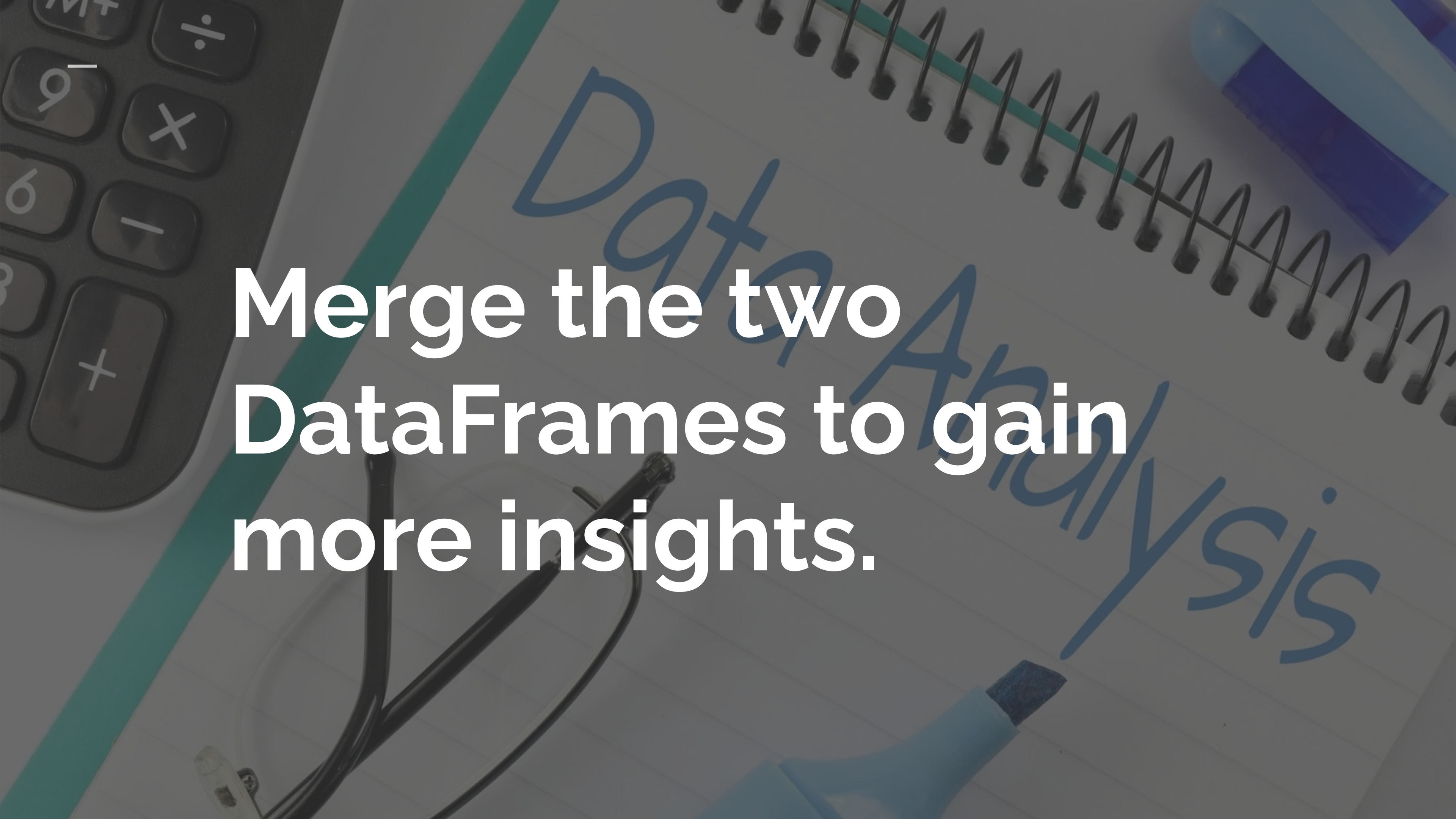
- People who are in a civil marriage, separated, or unmarried, and have only a lower secondary education, are at a higher risk of defaulting on their loans.



Multivariate Analysis

- We observed that individuals who took loans between 500k and 600k are more likely to default. The graph also shows that people in a civil marriage or those who are single have a higher risk of defaulting.





Merge the two DataFrames to gain more insights.

NAME_CONTRACT_STATUS	TARGET	
Approved	0	92.411345
	1	7.588655
Canceled	0	90.826431
	1	9.173569
Refused	0	88.003586
	1	11.996414
Unused offer	0	91.748276
	1	8.251724

92% of approved loans are repaid, and loans that were previously canceled or refused have also been repaid.

Conclusion: Clients with Payment Difficulties

Loan Amounts: Individuals who took loans between 500k and 600k have a higher likelihood of becoming defaulters.

Income Level: Clients with an income below 200k are more likely to default on their loans.

Age Group: Individuals in the 20 to 30 age group are less likely to repay their loans.

Employment Status: Unemployed individuals and those on maternity leave are more likely to default on their loans.

Education Level: People with only a lower secondary education have a higher risk of defaulting.

Family Size: Clients with a family size of more than seven are more likely to default.

Marital Status: Those in a civil marriage, separated, or unmarried with lower secondary education are at a higher risk of defaulting.

Loan Type: Cash loans have a higher default rate compared to revolving loans.

Previous Loan History: Individuals who have taken loans between 5 lakh and 6 lakh are more likely to default.

Loan Status: Those with previously canceled or refused loans that were later approved still show a risk of default.

Conclusion: Clients with a Higher Likelihood of Repaying Loans

Income Level: Clients with an income between 700k and 800k have a higher repayment rate.

Age Group: Individuals over the age of 50 are more likely to repay their loans.

Employment Status: Business owners and students have higher repayment rates compared to others.

Home Ownership: People who own their homes or apartments are more likely to repay their loans.

Education Level: Those with academic degrees are more likely to repay their loans compared to others.

Family Size: Clients with fewer children are more likely to repay their loans.

Loan Type: Despite a higher default rate, cash loans tend to have higher repayment amounts when they are repaid.

Marital Status: Married individuals with higher education are more likely to repay their loans.

Loan History: Clients who have successfully repaid previous loans are more likely to continue doing so.

Approved Loans: 92% of approved loans are repaid, indicating that approved clients generally have a strong repayment likelihood.


Thank You
Merci
Dank Je
Obrigado
Dyakuyu
Salamat
Arigato
Grazie
Kiitos
Juspaxar
Mamana
Spasibo
Multumesc
Mochchakkeram
Dankon
Vinaka
Asante
Maake
Chokrane
Kia Ora
Terma Kasih
Cam on ban
Raibh Maith Agat
Matur Nuwun
Welalin
Matondo
Salamat
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Ua Tsaug Rau Koj